**Keep in Mind**

**This is a practice test.** It mirrors the real Unit 1 test in length, format, and the ideas it covers — the numbers and the contexts are the only things that differ. Work every problem with no notes, then check yourself against the key. Write each final answer in its box. Only what is inside a box is graded. **80 points total.**

## Part A — Vocabulary (10 pts)

Write the letter of the best description in the box next to each term.

1. Function [C]
2. Domain [G]
3. Range [A]
4. Zero of a function [E]
5. Concave up [I]
6. Average rate of change [H]
7. Parent function [F]
8. Composition [B]
9. One-to-one [J]
10. Inverse function [D]

**(A)** The set of every output the rule actually produces.

**(B)** Feeding the output of one rule into another: $f(g(x))$.

**(C)** A rule that pairs each input with exactly one output.

**(D)** The rule that gives back the question: if $f(a) = b$, it sends $b$ to $a$.

**(E)** An input whose output is 0.

**(F)** The simplest member of a family, before any shift or stretch.

**(G)** The set of every input the rule is allowed to take.

**(H)** The one steady rate that would give the same change in output over the same run.

**(I)** Bending upward — the rate of change is increasing.

**(J)** No two different inputs share an output.

## Part B — Multiple Choice (12 pts)

Write the letter of the best answer in its Letter box at the right.

1. The statement $f(3) = 7$ tells you that

   (A) $f(7) = 3$ (B) 7 is an input of $f$ (C) $(3, 7)$ is on the graph of $f$ (D) $f$ is increasing at $x = 3$ **C**

2. On an interval where $f$ is decreasing and concave up, the outputs are

   (A) falling faster and faster (B) rising more and more slowly (C) changing at a constant rate (D) falling, but more and more slowly *Letter:* **D**

3. A table of $f$ has equally spaced inputs and **constant second** differences. The best model is

   (A) linear (B) quadratic (C) exponential (D) none of these *Letter:* **B**

4. The graph of $g(x) = f(x + 3)$ is the graph of $f$ shifted

   (A) 3 left (B) 3 right (C) 3 up (D) 3 down *Letter:* **A**

5. If $f(x) = x^2$ and $g(x) = x - 1$, then $(f \circ g)(4)$ equals

   (A) 15 (B) 16 (C) 9 (D) 3 *Letter:* **C**

6. A function has an inverse that is also a function exactly when the function is

   (A) increasing somewhere (B) concave up (C) defined at 0 (D) one-to-one *Letter:* **D**

## Part C — Short Answer & Computation (40 pts, 5 each)

Show your work in the space provided.

1. Let $f(x) = 2x^2 - 5$.

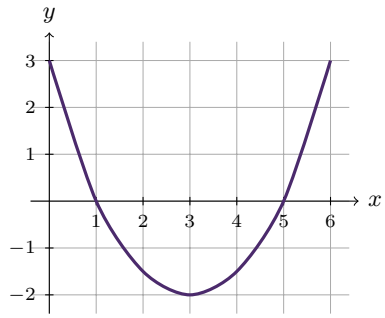
$$\begin{aligned} f(3) &= 2(3)^2 - 5 \\ &= 13 \\ f(-2) &= 2(-2)^2 - 5 \\ &= 3 \\ 2x^2 - 5 &= 3 \\ x^2 &= 4 \\ x &= \pm 2 \end{aligned}$$

(a) $f(3) =$ 13 **(b)** $f(-2) =$ 3

(c) Solve $f(x) = 3$. There are two answers: $x = 2, \ -2$

(d) Which of (a)–(c) is a *solve* question rather than an *evaluate* question? **(c)**

2. The graph of $y = f(x)$ is shown below.




(a) On what interval is $f$ decreasing? $0 < x < 3$

(b) Concave up or concave down on the whole picture? **Concave up**

(c) List the zeros of $f$: $x = 1$ **and** $x = 5$

(d) $f(3) =$ $-2$ **(e)** Range on the window shown: $-2 \le y \le 3$

3. A tank is draining. Its volume $V$ (gallons) after $t$ minutes:

| $t$ (min) | 0 | 4 | 8 | 12 |
|---|---|---|---|---|
| $V$ (gal) | 180 | 148 | 124 | 108 |

$$\frac{V(8) - V(0)}{8 - 0} = \frac{124 - 180}{8} = -7$$

(a) Average rate of change of $V$ from $t = 0$ to $t = 8$: $-7$, with units **gallons per minute**.

(b) In one sentence, what does that number mean for the tank?

Over those first 8 minutes the tank lost water at an average of 7 gallons a minute.

(c) Is the tank draining faster or slower as time goes on? **Slower** Justify with the numbers.

The drop over each 4-minute block shrinks: $-32$, then $-24$, then $-16$ gallons.

4. The table below has equally spaced inputs. Fill in the difference rows, then name the function type.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $g(x)$ | 2 | 5 | 12 | 23 | 38 |
| 1st differences | — | 3 | 7 | 11 | 15 |
| 2nd differences | — | — | 4 | 4 | 4 |

Function type: **quadratic** Explain how the table shows it.

Because the 2nd differences are constant.

5. The parent function is $f(x) = \sqrt{x}$, with domain $[0, \infty)$ and range $[0, \infty)$. Let $g(x) = -2f(x-1) + 4$.

(a) Write $g$ in terms of $x$: $g(x) =$ $-2\sqrt{x-1}+4$

(b) The change *inside* the parentheses moves the graph **1 unit right**

(c) The changes *outside*, in order: **stretch by 2, reflect over the $x$-axis, up 4**

(d) Domain of $g$: $[1, \infty)$ Range of $g$: $(-\infty, 4]$

(e) The parent's landmark point $(0, 0)$ lands at $\boxed{(1, 4)}$ on $g$.

6. Two one-to-one functions are given by the table below.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $f(x)$ | 5 | 2 | 7 | 1 | 3 |
| $g(x)$ | 3 | 0 | 4 | 2 | 1 |

(a) The handoff in $f(g(1))$ is $\boxed{0}$, so $f(g(1)) = \boxed{5}$

(b) $g(f(3)) = \boxed{0}$

(c) $f^{-1}(7) = \boxed{2}$ because $f(\boxed{2}) = 7$.

(d) How can you tell from the table that $f$ is one-to-one?

No output repeats in the $f$ row, so no two inputs share an output.

(e) Why does $g(f(0))$ have no value from this table?

$f(0) = 5$, and 5 is not one of the inputs $g$ accepts here.

7. Let $f(x) = 3x + 1$ and $g(x) = x^2 - 2$.

$$
\begin{aligned}
g(2) &= 4 - 2 \\
&= 2 \\
f(g(2)) &= 3(2) + 1 \\
&= 7 \\
f(2) &= 7 \\
g(f(2)) &= 7^2 - 2 \\
&= 47 \\
(f \circ g)(x) &= 3(x^2 - 2) + 1 \\
&= 3x^2 - 5
\end{aligned}
$$

(a) $f(g(2)) =$ 7 **(b)** $g(f(2)) =$ 47

(c) $(f \circ g)(x) =$ $3x^2 - 5$

(d) What do (a) and (b) show about the order of a composition?

Order matters: $7 \neq 47$, so $f(g(x))$ and $g(f(x))$ are different functions.

8. Let $f(x) = \dfrac{x-5}{2}$.

$$
\begin{aligned}
y &= \frac{x-5}{2} \\
x &= \frac{y-5}{2} \\
2x &= y - 5 \\
y &= 2x + 5 \\
f(f^{-1}(x)) &= \frac{(2x+5) - 5}{2} \\
&= x
\end{aligned}
$$

(a) Find $f^{-1}$ by swapping and solving: $f^{-1}(x) =$ $2x + 5$

(b) Verify: $f\big(f^{-1}(x)\big) =$ $x$

(c) $f(11) = 3$, so $f^{-1}(3) =$ 11

## Part D — Extended Response (18 pts, 9 each)

Explain your reasoning in full sentences.

1. A rectangular dog run is built against a barn wall, so only **three** sides need fencing. The builder has 60 feet of fence. Let $x$ be the length of each of the two sides perpendicular to the barn.

$$\begin{aligned} A(x) &= x(60 - 2x) \\ &= 60x - 2x^2 \\ A(5) &= 5(50) \\ &= 250 \\ A(10) &= 10(40) \\ &= 400 \\ \frac{A(10) - A(5)}{10 - 5} &= \frac{400 - 250}{5} \\ &= 30 \end{aligned}$$

(a) Which quantity is the *question* the model answers — its input? **$x$, the perpendicular side**

(b) The side parallel to the barn has length $60 - 2x$, so $A(x) = 60x - 2x^2$

(c) What type of function is $A$, and how do you know?

Quadratic: the highest power of $x$ is 2, so second differences are constant.

(d) Domain the context forces: $0 < x < 30$ Why?

A side must be positive, and the barn side $60 - 2x$ must be positive too.

(e) Average rate of change of $A$ from $x = 5$ to $x = 10$: 30, with units **ft$^2$ per foot of $x$**.

(f) State one assumption this model rests on.

No fence runs along the barn, and none is lost to a gate or an overlap.

2. Let $f(x) = x^2$ with domain all real numbers.

(a) Explain why $f$ has no inverse function on that domain. Give a counterexample pair of inputs.

Not one-to-one: $f(2) = f(-2) = 4$, so reversing sends 4 to two outputs.

(b) Give a domain restriction that fixes the problem: $x \geq 0$ Then $f^{-1}(x) =$ $\sqrt{x}$

(c) Domain of that inverse: $[0, \infty)$ Range: $[0, \infty)$

(d) Explain how the graph of a function and the graph of its inverse are related, and why.

They are reflections of each other over the line $y = x$. The inverse trades input for output, so $(a, b)$ on $f$ becomes $(b, a)$ on $f^{-1}$.